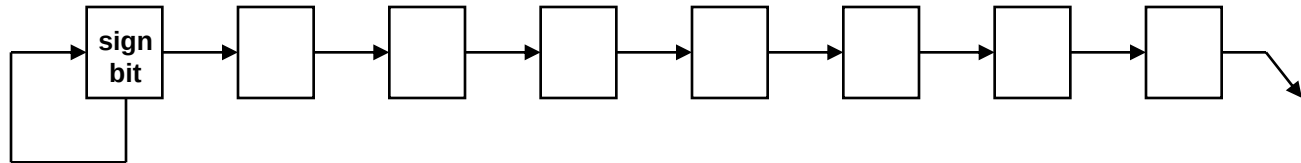
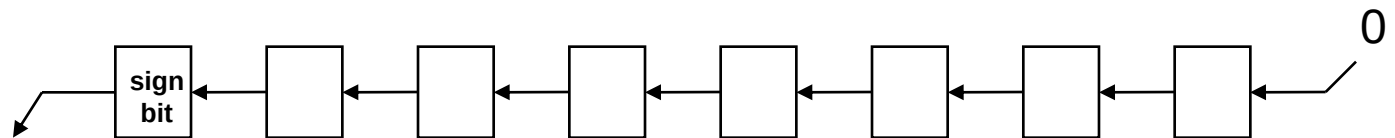


ARITHMETIC SHIFT

- An arithmetic shift is meant for signed binary numbers (integer)
 - An arithmetic left shift multiplies a signed number by two
 - An arithmetic right shift divides a signed number by two
 - The main distinction of an arithmetic shift is that it must keep the sign of the number the same as it performs the multiplication or division
-
- A right arithmetic shift operation:

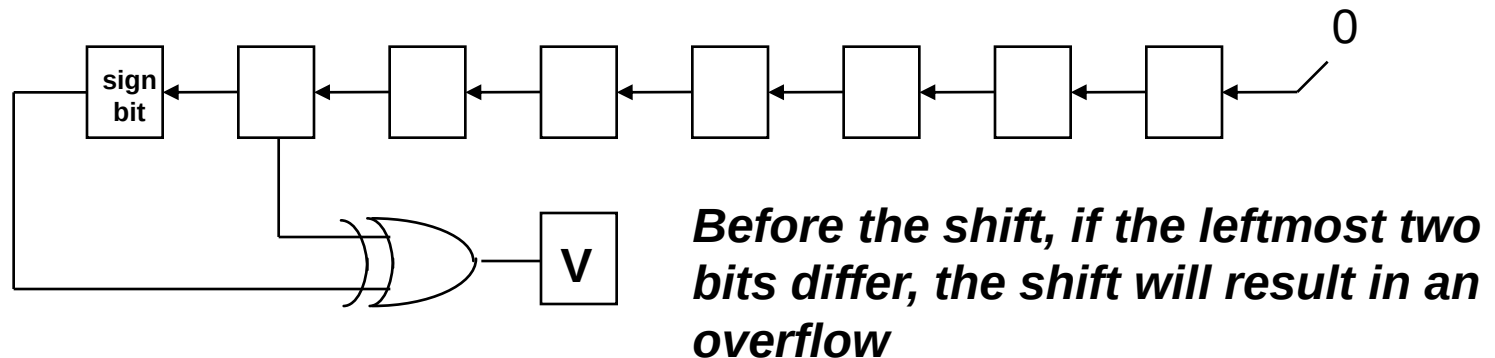


- A left arithmetic shift operation:



ARITHMETIC SHIFT

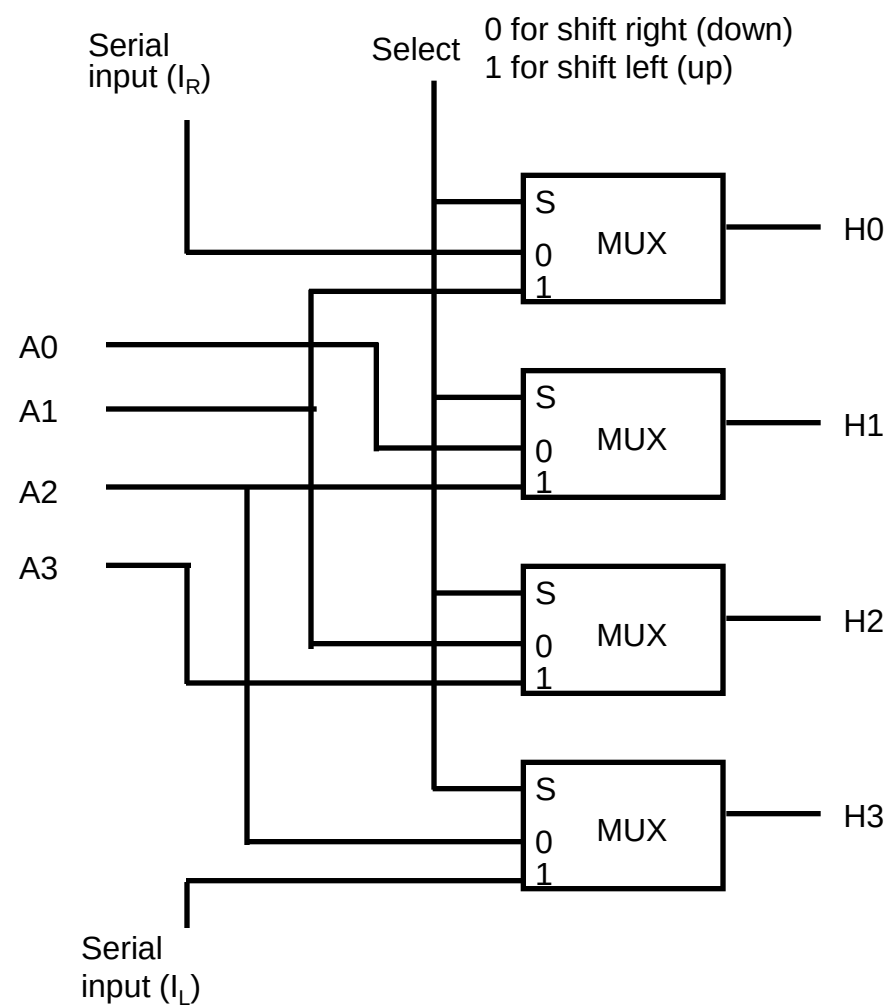
- An left arithmetic shift operation must be checked for the overflow



- In a RTL, the following notation is used
 - *ashl* for an arithmetic shift left
 - *ashr* for an arithmetic shift right
 - Examples:
 - » $R2 \leftarrow ashr\ R2$
 - » $R3 \leftarrow ashl\ R3$



HARDWARE IMPLEMENTATION OF SHIFT MICROOPERATIONS



Function table

Select	Output			
	H ₀	H ₁	H ₂	H ₃
0	I _R	A ₀	A ₁	A ₂
1	A ₁	A ₂	A ₃	I _L



ARITHMETIC LOGIC SHIFT UNIT

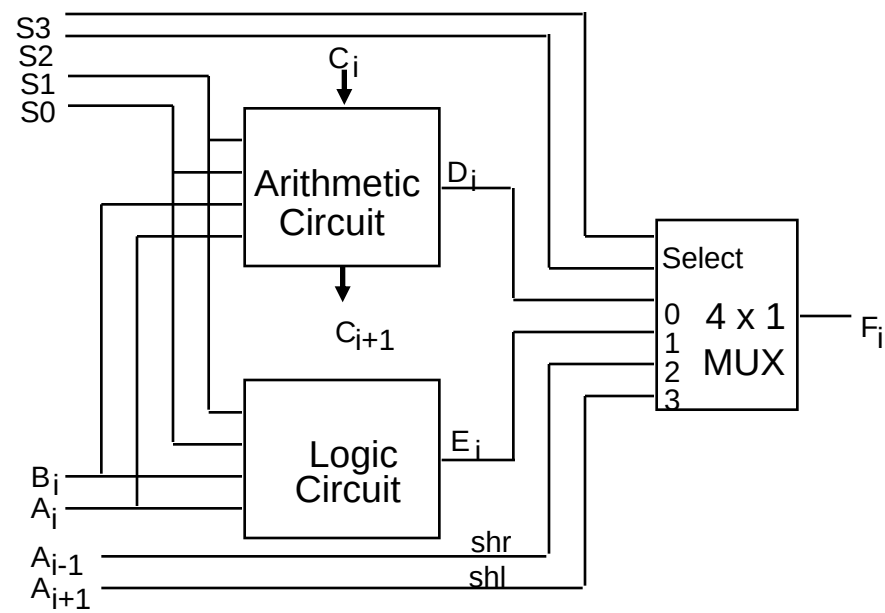


TABLE 4-8 Function Table for Arithmetic Logic Shift Unit

Operation select					Operation	Function
S ₃	S ₂	S ₁	S ₀	C _{in}		
0	0	0	0	0	$F = A$	Transfer A
0	0	0	0	1	$F = A + 1$	Increment A
0	0	0	1	0	$F = A + B$	Addition
0	0	0	1	1	$F = A + B + 1$	Add with carry
0	0	1	0	0	$F = A + \overline{B}$	Subtract with borrow
0	0	1	0	1	$F = A + \overline{B} + 1$	Subtraction
0	0	1	1	0	$F = A - 1$	Decrement A
0	0	1	1	1	$F = A$	Transfer A
0	1	0	0	×	$F = A \wedge B$	AND
0	1	0	1	×	$F = A \vee B$	OR
0	1	1	0	×	$F = A \oplus B$	XOR
0	1	1	1	×	$F = \overline{A}$	Complement A
1	0	×	×	×	$F = \text{shr } A$	Shift right A into F
1	1	×	×	×	$F = \text{shl } A$	Shift left A into F

